

CRUX: Caption-Fused and Re-ranked News Image Retrieval with LLM Query Expansion

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Abstract—This paper presents a modular retrieval pipeline for the NewsImages 2026 challenge at MediaEval, where the task is to recommend fitting images for news article titles. Our system starts from a strong CLIP baseline and incrementally integrates three components: (i) LLM-based query expansion (LLM-QE), (ii) caption-aware embedding fusion using BLIP2 captions, and (iii) a second-stage CLIP re-ranker. We evaluate all variants on the official evaluation split using Recall@1/5/10 and MRR from our internal benchmark script. Results show that CLIP re-ranking provides the most consistent gains over baseline (Recall@10: 0.564 $\rightarrow$ 0.573, MRR: 0.334 $\rightarrow$ 0.340), while query expansion adds small but stable improvements in retrieval recall. Caption fusion alone underperforms due to caption noise/mismatch, but remains useful as an auxiliary signal in a hybrid setting. The main contribution is an adaptive, fully reproducible retrieval pipeline that combines semantic query rewriting and multi-view representation scoring with lightweight heuristics, while keeping competitive accuracy and practical runtime.

Index Terms—News image recommendation, MediaEval, CLIP retrieval, query expansion, multimodal fusion, re-ranking

I. INTRODUCTION

Selecting an appropriate image for a news article is a challenging multimodal retrieval problem because article titles are short, ambiguous, and often context-dependent. The NewsImages task at MediaEval 2026 requires participants to recommend fitting images for news content under realistic constraints, emphasizing semantic relevance rather than exact lexical overlap.

In this work, we focus on a retrieval-centric approach and study how far a carefully engineered CLIP pipeline can be pushed without supervised fine-tuning. We begin with a zero-shot CLIP retrieval baseline and add three orthogonal enhancements: (1) LLM-based query expansion to enrich sparse article titles, (2) image-caption fusion to inject additional textual grounding through BLIP2-generated captions, and (3) candidate re-ranking with a higher-resolution CLIP model for improved top-k ordering.

Our goal is not only to maximize ranking quality but also to provide an interpretable and modular workflow for MediaEval submissions. Each component can be enabled/disabled independently, making the system practical for ablation, reproducibility, and future extension.

II. RELATED WORK

A. News Image Retrieval

The NewsImages benchmark at MediaEval has been an active testbed for image-to-article matching. Early editions focused primarily on comparing image retrieval against generative approaches [2], while more recent editions have moved toward hybrid systems that blend both paradigms [1]. Supporting work has also examined how humans perceive the relevance between news texts and images, highlighting that semantic alignment is often subtle and context-sensitive [3]. Our work builds on this retrieval-centric line, studying how far a zero-shot pipeline can be taken without task-specific fine-tuning.

B. CLIP for Zero-Shot Image Retrieval

Contrastive Language-Image Pre-training (CLIP) [4] trains a dual-encoder model on large-scale image-text pairs using a contrastive objective that brings matching pairs closer in embedding space while pushing non-matching pairs apart. Because both modalities are projected into the same space, CLIP supports zero-shot retrieval by directly computing cosine similarity between a text query and image embeddings. This makes it a natural baseline for tasks like NewsImages, where queries are short article titles and the image corpus is fixed. CLIP’s strong zero-shot generalization, particularly with larger vision backbones such as ViT-L/14 and ViT-L/14@336px, has made it the dominant starting point for retrieval systems in the vision-language community [7].

C. Vision-Language Models for Image Captioning

Generating descriptive captions for images provides an additional textual view of visual content that can complement embedding-based retrieval. BLIP [9] introduced a unified vision-language pre-training framework with a multimodal mixture of encoder-decoder (MED) that jointly supports understanding and generation tasks. Crucially, BLIP also proposed CapFilt, a method for bootstrapping noisy web-collected image-text pairs using a captioner and a filter, which substantially improved downstream performance. Building on this, BLIP-2 [5] proposed a more compute-efficient approach by keeping both the image encoder and a large language model frozen, and instead training a lightweight Querying

Transformer (Q-Former) to bridge the modality gap. BLIP-2 achieves strong captioning quality with significantly fewer trainable parameters than prior methods. In our pipeline, we use precomputed BLIP-2 captions as an auxiliary textual signal on the image side, encoding them in the same CLIP text space to enrich image representations.

D. LLM-Based Query Expansion

Short text queries, such as news headlines, often lack sufficient context for a retrieval model to identify the intended image. Query expansion addresses this by rewriting or augmenting the original query with additional descriptive content. Zhu et al. [8] demonstrated that large language models (LLMs) and vision-language models (VLMs) can be used together in a multi-turn interactive retrieval system: relevant images are captioned by a VLM, and an LLM then rewrites the query based on those captions, leading to measurable gains in recall across interaction turns. Their work also showed that a Chain-of-Thought prompting approach for query summarization outperforms simpler expansion strategies. Inspired by this, our system uses a local LLM (Phi-3 via Ollama) to expand article titles with semantically consistent context, while applying a similarity gate to prevent harmful drift from over-expansion.

E. Two-Stage Re-ranking

A common limitation of single-stage retrieval is that the top-ranked results may not be optimally ordered, especially when the initial embeddings cannot reason jointly over query and candidate. Two-stage pipelines address this by first retrieving a small candidate pool with a fast embedding-based model, then re-scoring those candidates with a more expressive model. Liu et al. [6] proposed a dual-encoder re-ranking architecture for composed image retrieval that explicitly attends to query-candidate triplets, yielding consistent improvements over single-stage approaches across multiple benchmarks. This confirms that re-ranking is particularly effective for separating visually similar hard negatives. In our system, we adopt a lighter version of this idea: a higher-resolution CLIP model (ViT-L/14@336px) re-scores the top-100 candidates from the initial ViT-L/14 retrieval, improving top-rank precision without requiring a separate architecture.

F. Summary

Our pipeline draws on all four of the above lines of work. CLIP provides the zero-shot retrieval backbone; BLIP-2 captions enrich image-side representations; LLM-based query rewriting enriches the query side; and a second-stage CLIP re-ranker refines the final ordering. The key contribution is combining these components in a modular, gated fashion tailored to the short-query, noisy-caption conditions of the NewsImages setting.

III. TASK AND DATA

We follow the NewsImages 2026 setup, using the official evaluation subset with article titles and associated ground-truth images from the MediaEval release. The broader challenge

includes around 8,500 training articles and approximately 14000 evaluation and 800 test items (English news from MIND and historical archives), but our reported offline metrics are computed on the evaluation split where image IDs are available for benchmarking [1].

For each article title, the system retrieves ranked image candidates from the evaluation image pool. We report Recall@1, Recall@5, Recall@10, and Mean Reciprocal Rank (MRR), which reflect both hit quality and rank sensitivity.

IV. METHOD

Our implementation is organized as independent scripts that share a common utility and model layer. This section summarizes the retrieval architecture.

A. Baseline: CLIP Retrieval

The baseline encodes article titles and candidate images with CLIP ViT-L/14 [4]. Image embeddings are cached to disk for efficiency, and all embeddings are L2-normalized. Ranking is computed by cosine similarity in embedding space.

B. LLM-Based Query Expansion (LLM-QE)

We use a local Phi-3 model via Ollama to rewrite article titles into slightly expanded queries under strict constraints: no new facts, no fabricated events, and short output. The expansion is cached in JSON to ensure deterministic reuse across runs. This approach is motivated by recent work showing that LLM-generated query rewrites improve retrieval recall, particularly for short or ambiguous queries [8].

Given original query embedding q and expanded query embedding q_{exp} , we compute their similarity and apply adaptive interpolation only when semantic alignment is sufficiently high. This prevents harmful drift from over-expansion.

C. Caption Fusion

To enrich image-side semantics, we use BLIP-2 captions [5] (precomputed in an auxiliary CSV) and encode them in the same CLIP text space. BLIP-2 generates high-quality image descriptions by bridging a frozen image encoder and a frozen large language model through a lightweight Q-Former, making it well-suited for producing compact and informative captions [9]. For each image, we fuse image and caption embeddings only if their mutual similarity exceeds a threshold; otherwise, we keep the original image embedding. This conditional fusion avoids forcing noisy captions into the representation.

D. CLIP Re-ranking

A two-stage ranking strategy is used: first-stage retrieval with ViT-L/14 retrieves top- k candidates ($k=100$), then a second CLIP model (ViT-L/14@336px) re-scores this candidate set against the original query for better ordering at the top ranks. Two-stage re-ranking has been shown to be highly effective at separating visually similar hard negatives that the initial retrieval stage cannot distinguish [6].

E. Hybrid Pipeline

Our final hybrid variant combines the above components:

- adaptive query expansion on the query side,
- conditional image-caption fusion on the index side,
- second-stage CLIP re-ranking on candidates.

The design is intentionally modular and can be tuned through a small set of transparent hyperparameters (fusion beta, QE beta, similarity thresholds, top-k re-ranking depth).

V. EXPERIMENTAL SETUP

Hardware acceleration is selected automatically (CUDA, MPS, or CPU). Batch encoding is performed in fixed-size chunks and cached in the artifacts directory to reduce repeated compute. All runs use the same evaluation split and common metric code for fair comparison.

Compared systems:

- 1) CLIP Retrieval (baseline)
- 2) CLIP + Re-ranker
- 3) CLIP + LLM-QE
- 4) CLIP + LLM-QE + Re-ranker
- 5) CLIP + Caption Fusion
- 6) CLIP + Caption Fusion + Re-ranker
- 7) Hybrid (LLM-QE + Caption Fusion + Re-ranker)

VI. RESULTS AND DISCUSSION

Table I summarizes the logged results.

TABLE I
OFFLINE EVALUATION ON NEWSIMAGES EVALUATION SPLIT

System	R@1	R@5	R@10	MRR
CLIP	0.239	0.462	0.564	0.334
CLIP+Re-ranker	0.244	0.469	0.573	0.340
CLIP+LLM-QE	0.241	0.465	0.570	0.337
CLIP+LLM-QE+Re-ranker	0.244	0.469	0.573	0.340
CLIP+Caption Fusion	0.176	0.353	0.442	0.252
CLIP+Caption Fusion+Re-ranker	0.241	0.455	0.549	0.332
CRUX	0.240	0.452	0.544	0.330

Key observations:

First, CLIP re-ranking is the strongest single improvement. It consistently improves all metrics over the plain baseline, indicating that better top-k ordering is more impactful than large architectural changes in this setting.

Second, LLM-QE yields modest gains over baseline and appears complementary to re-ranking, but its contribution is smaller than re-ranking alone. This is expected because many article titles are already informative enough for CLIP, limiting expansion headroom.

Third, caption fusion alone performs substantially worse than baseline. This suggests that caption quality/domain mismatch can introduce noise in news retrieval, particularly when captions over-describe visual details not aligned with headline semantics.

Fourth, adding re-ranking after caption fusion recovers most of the lost performance, showing that multi-stage ranking can compensate for noisier first-stage representations.

Finally, the full hybrid model does not surpass the best re-ranking variants in our current configuration. This indicates non-trivial interaction effects between QE and fusion; stronger gains likely require better calibration of thresholds and per-query adaptive weighting.

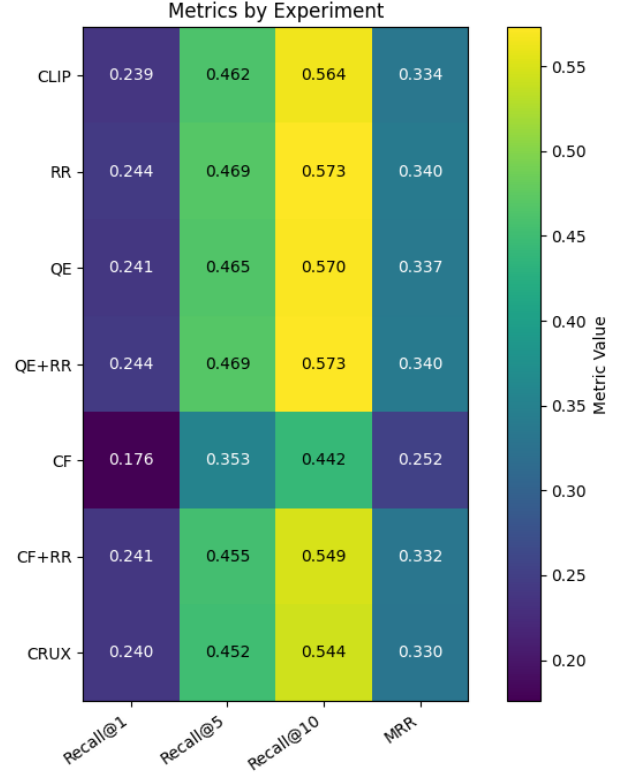


Fig. 1. Performance Heatmap

VII. NOVELTY AND CONTRIBUTION

The novelty of this work is not a new pretrained backbone, but a practical algorithmic integration tailored to NewsImages:

1) Adaptive dual-side conditioning:

- Query-side conditioning through similarity-gated LLM expansion.
- Image-side conditioning through similarity-gated caption fusion.

2) Modular three-stage retrieval design:

- initial semantic retrieval,
- optional semantic enrichment,
- high-resolution reranking for final ordering.

3) Reproducible engineering for challenge workflows:

- cached embeddings and cached expansions,
- script-level ablation support,
- consistent metric logging across all variants.

This combination provides a transparent and extensible framework for short-paper challenge submissions, where reproducibility and comparative analysis are essential.

VIII. LIMITATIONS AND FUTURE WORK

Our evaluation is offline and based on retrieval metrics against known evaluation images; final challenge outcomes depend on human judgments in the online event. Also, caption quality is inherited from upstream captioning and can propagate noise. Future work will include:

- confidence-calibrated QE weights per query type,
- improved caption filtering/denoising,
- diversification-aware ranking to avoid visually redundant outputs,
- optional retrieval-generation hybrids for difficult or low-confidence articles.

IX. CONCLUSION

We presented a modular retrieval pipeline for NewsImages 2026 that combines CLIP retrieval, LLM-assisted query expansion, caption-aware fusion, and CLIP re-ranking. Among all tested components, re-ranking provides the most reliable gains, while query expansion offers smaller complementary improvements. Caption fusion is sensitive to caption quality and requires careful gating. Overall, the study shows that lightweight adaptive controls and strong two-stage ranking can deliver competitive performance for news image recommendation without expensive fine-tuning.

REFERENCES

- [1] L. Heitz et al., “NewsImages in MediaEval 2026 - Automated Image Recommendations with Retrieval and Generation Techniques for News Articles Thumbnails,” Working Notes Proceedings of the MediaEval 2026 Workshop, 2026.
- [2] L. Heitz et al., “NewsImages in MediaEval 2025 - Comparing Image Retrieval and Generation for News Articles,” Working Notes Proceedings of the MediaEval 2025 Workshop, 2025.
- [3] L. Heitz, L. Rossetto, and A. Bernstein, “An Empirical Exploration of Perceived Similarity between News Article Texts and Images,” Working Notes Proceedings of the MediaEval 2023 Workshop, 2024.
- [4] A. Radford, J. W. Kim, C. Hallacy, A. Ramesh, G. Goh, S. Agarwal, G. Sastry, A. Askell, P. Mishkin, J. Clark, G. Krueger, and I. Sutskever, “Learning Transferable Visual Models From Natural Language Supervision,” in *Proc. ICML*, 2021.
- [5] J. Li, D. Li, S. Savarese, and S. Hoi, “BLIP-2: Bootstrapping Language-Image Pre-training with Frozen Image Encoders and Large Language Models,” in *Proc. ICML*, 2023.
- [6] Z. Liu, W. Sun, D. Teney, and S. Gould, “Candidate Set Re-ranking for Composed Image Retrieval with Dual Multi-modal Encoder,” *arXiv preprint arXiv:2305.16304*, 2024.
- [7] H. S and N. K. Lahajjal, “Enhancing Image Retrieval: A Comprehensive Study on Photo Search using the CLIP Model,” *BMS College of Engineering*, 2023.
- [8] H. Zhu, J.-H. Huang, S. Rudinac, and E. Kanoulas, “Enhancing Interactive Image Retrieval With Query Rewriting Using Large Language Models and Vision Language Models,” in *Proc. ACM ICMR*, 2024.
- [9] J. Li, D. Li, C. Xiong, and S. Hoi, “BLIP: Bootstrapping Language-Image Pre-training for Unified Vision-Language Understanding and Generation,” in *Proc. ICML*, 2022.